Standard Operating Procedure (SOP)

VIPHERAID – Animal Rescue and Shelter Management System

1. Purpose

The purpose of the VipherAid system is to automate and digitize the process of animal rescue reporting and shelter coordination. The system enables citizens to quickly report injured or stray animals using minimal input, ensuring faster response and improved animal welfare.

The platform uses location-based services and image uploads to provide accurate and real-time information to rescuers. It also integrates a transparent donation system that connects users directly to multiple shelters for fundraising support.

The system eliminates delays caused by manual reporting methods and provides a centralized, web-based solution for managing rescue operations, shelter allocation, and donation access.

2. Scope

This SOP applies to:

- **Citizens** — Individuals who report injured or stray animals by uploading photos and enabling location access
- **Rescuers** — Personnel or organizations responsible for responding to reports and performing rescue operations
- **Shelters** — Facilities that receive and care for rescued animals and manage their capacity
- **Donors** — Users who contribute financially to shelters through transparent redirection links

The system is deployed for daily use in animal rescue scenarios and shelter coordination activities.

3. Definitions

Term	Definition
VipherAid	Web-based Animal Rescue and Shelter Management System
Report	A record created when a user submits an animal complaint
Shelter	A facility that houses and treats rescued animals
Rescuer	Individual or group responsible for animal rescue
Location Tracking	Automatic detection of user's geographic position
Photo Upload	Capturing and submitting an image of the animal
Donation Redirection	Feature that redirects users to multiple shelters for funding
Report Status	Current state of rescue (Pending, In Progress, Completed)

4. Roles and Responsibilities

Citizens

- Access the web application
- Upload an image of the animal
- Allow location access for accurate reporting
- Submit reports with correct information

Rescuers

- Monitor incoming reports in real time
- Accept and respond to assigned reports
- Perform rescue operations
- Transport animals safely to shelters

- Update report status during each stage

Shelters

- Maintain records of available capacity
- Accept rescued animals
- Provide necessary care and treatment
- Update availability and intake status

Donors

- Access donation page
- View multiple shelter options
- Choose preferred shelter for donation
- Support transparent fundraising

5. System Requirements

Hardware

- Smartphone, tablet, or computer
- Internet connection
- GPS-enabled device for location tracking

Software

- Web browser (Google Chrome, Microsoft Edge, Mozilla Firefox)
- VipherAid Web Application
- Backend framework (Flask or equivalent)

- Database system (SQLite or similar)

Environment Configuration

- Stable internet connectivity
- Location services enabled in browser/device
- Image upload permissions enabled

6. Procedure

Step 1 — System Access

1. Open the VipherAid application in a web browser
2. Allow location permission when prompted
3. The system loads the home page

If permission is denied, the system prompts the user again to enable location services.

Step 2 — Animal Reporting (Report Page)

On the report page, the user performs the following:

1. Upload an image of the animal
2. The system automatically captures the current location
3. Click the **Submit** button

The system then:

- Validates the uploaded image
- Captures and stores location data
- Creates a new report entry

- Assigns default status as **"Pending"**
- Displays confirmation message

Note:
No additional fields are required to simplify and speed up reporting.

Step 3 — Report Monitoring and Acceptance

1. Rescuers access the system dashboard
2. View all active reports
3. Select a report to respond

The system then:

- Assigns the selected report to the rescuer
- Updates status to **"In Progress"**
- Prevents duplicate assignment

Step 4 — Rescue Operation

1. Rescuer travels to the reported location
2. Identifies the animal using the uploaded image
3. Performs necessary rescue actions

The system allows:

- Status updates during rescue
- Tracking of rescue progress

Step 5 — Shelter Allocation

1. System identifies nearby shelters
2. Checks available capacity

The system then:

- Selects the most suitable shelter
- Updates shelter capacity records
- Links the report to the selected shelter

Step 6 — Animal Admission

1. Animal is transported to the assigned shelter
2. Shelter confirms intake

The system then:

- Updates report status to **"Completed"**
- Stores final rescue record
- Logs all activities for tracking

Step 7 — Donation Process (Transparent Fundraising)

1. User navigates to the donation page
2. The system displays multiple shelters

The system then:

- Redirects users to different shelter donation links
- Provides transparency by showing multiple options
- Allows users to choose their preferred shelter

Step 8 — Viewing Reports and Status

Users can:

- View submitted reports
- Track current status

Rescuers can:

- Filter reports based on status
- Monitor ongoing operations

7. Database Structure

The system uses a structured database with the following tables:

Table	Purpose	Key Fields
reports	Stores animal reports	id, image_url, location, status
shelters	Stores shelter details	id, name, capacity, availability
rescuers	Stores rescuer information	id, name, contact
donation	Tracks redirection links	shelter_name, donation_link

8. Data Management and Security

- Secure session-based access
- Data stored in structured database
- Role-based usage without administrative complexity
- Safe handling of uploaded images
- Location data used strictly for rescue purposes

- Regular monitoring of system performance

9. Error Handling

- Missing image → Prompt user to upload
- Location disabled → Request permission
- Invalid data → Display clear error message
- Duplicate report handling
- Database errors handled safely without system crash

10. Deployment and Maintenance

Deployment

- Hosted on local server or cloud platform
- Accessible through web browsers
- Backend connected to database

Maintenance Tasks

- Monitor incoming reports
- Update shelter information
- Ensure system uptime
- Perform periodic database backups
- Check donation links regularly

11. Benefits of the System

- Simplified reporting (only photo and location)
- Faster and efficient rescue operations
- Transparent donation system with multiple shelters
- Real-time status tracking
- Reduced manual effort
- Improved coordination between rescuers and shelters
- Centralized and scalable solution

12. Conclusion

VipherAid offers a reliable and efficient system for managing animal rescue operations and shelter coordination. By minimizing user input requirements and providing transparent donation access, the system enhances usability, accountability, and overall effectiveness.

The integration of real-time reporting, structured workflows, and shelter management ensures improved response time and better care for animals in need.